

Problem 1: Sample of birth weights (g) of live-born infants at a hospital during a 1-week period:

i	x_i	i	x_i	i	x_i	i	x_i
1	3265	6	3323	11	2581	16	2759
2	3260	7	3649	12	2841	17	3248
3	3245	8	3200	13	3609	18	3314
4	3484	9	3031	14	2838	19	3101
5	4146	10	2069	15	3541	20	2834

Calculate the arithmetic mean. If the first infant happened to be premature weighing 500 g rather than 3265 g, what would be the arithmetic mean?

Problem 2: Price per tonne of iron purchased by an industry for six months:

Month	Price per tonne (Rs.)	Iron purchased (tonnes)
January	42	25
February	51	35
March	50	31
April	40	47
May	60	48
June	54	50

Calculate the simple and weighted arithmetic average price per tonne of iron.

Problem 3: Calculate the median for the distribution in Problem 1.

Problem 4: Sample of admission white blood cell counts ($\times 1000$) for all patients entering a hospital on a given day:

i	x_i	i	x_i
1	7	6	3
2	35	7	10
3	5	8	12
4	9	9	8
5	8		

Calculate the mean and median white blood count. Will the mean and median values change if the second patient had a white blood count of 65,000 rather than 35,000?

Problem 5: Haemoglobin measurements (g/dl) from 48 intensive care patients

5.4	8.2	9.3	9.9	10.5	11.9
6.4	8.3	9.4	9.9	10.5	12.3
6.4	8.3	9.4	9.9	10.6	12.6
7.0	8.6	9.4	10.1	10.8	12.7
7.1	8.8	9.4	10.3	10.8	13.0
7.3	8.9	9.5	10.3	11.3	13.3
7.7	9.1	9.7	10.4	11.7	14.0
8.1	9.3	9.7	10.4	11.7	14.1

Calculate the mean and median for the above sample, and comment on the skewness of the distribution.

Problem 6: Biceps skin-fold thickness (mm) in patients with Crohn's disease:

1.8	2.8	4.2	6.2
2.2	3.2	4.4	6.6
2.4	3.6	4.8	7.0
2.5	3.8	5.6	10.0
2.8	4.0	6.0	10.4

Calculate the mean and median for the above sample, and comment on the skewness of the distribution.

Problem 7: Compute the mode of the distributions given in Problems 4 to 6.

Problem 8: Scores of two batsmen in 5 one-day matches:

Batsman A	Batsman B
30	2
33	21
35	31
37	58
40	63

Calculate the mean scores of the two batsmen, and comment on the consistency of their performances.

Problem 9: Two samples of cholesterol measurements (mg/dl) of a given person using two different measurement methods:

Method 1	Method 2
177	192
193	197
195	200
209	202
226	209

Calculate the mean and median for the above two measurements. Calculate the ranges for the two different methods, and compare their variability.

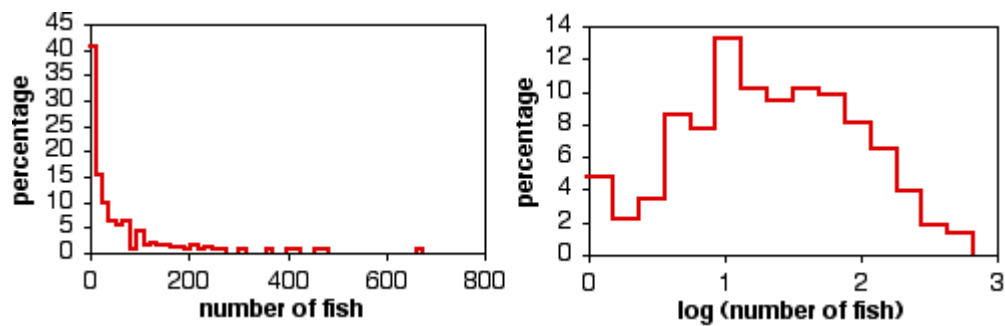
Problem 10: Calculate variance and standard deviation of the distribution given in Problem 8.

Problem 11: Calculate coefficient of variation from the following data:

Mean	s
80	12
0.50	0.20

Problem 12: The wages of workers are in rupees, and the consumption of chocolate in their families is in grammes. If the CV for wages is 10% and that for chocolate consumption is 25%, what can be concluded?

Problem 13: The first graph below depicts the abundance of the fish species *Umbra pygmaea* in streams in Maryland, USA. The second graph depicts the same data after applying log transformation.



Twelve data points from the fish abundance measurements are given below. Calculate the arithmetic mean and the mean obtained after log transformation.

38	2	50	6
1	13	9	4
13	20	28	43

Problem 14: Scores given by two judges to six contestants in a competition are given below:

	Judge A	Judge B
Contestant 1	24	17
Contestant 2	44	34
Contestant 3	25	16
Contestant 4	40	31
Contestant 5	39	43
Contestant 6	45	40

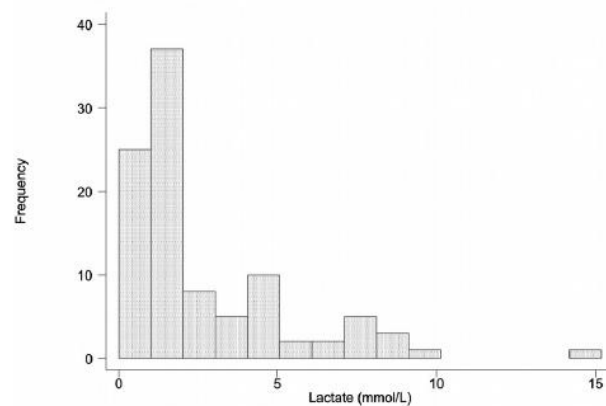
Normalize the scores given by the two judges to determine the winning contestant.

Problem 15: The test scores of students in a class test has a mean of 70 and a standard deviation of 12. What is the probable percentage of students who scored more than 85? Assume that the test scores are normally distributed.

Problem 16: Calculate the 95% reference range for haemoglobin measurements (g/dl) from 48 intensive care patients.

5.4	8.2	9.3	9.9	10.5	11.9
6.4	8.3	9.4	9.9	10.5	12.3
6.4	8.3	9.4	9.9	10.6	12.6
7.0	8.6	9.4	10.1	10.8	12.7
7.1	8.8	9.4	10.3	10.8	13.0
7.3	8.9	9.5	10.3	11.3	13.3
7.7	9.1	9.7	10.4	11.7	14.0
8.1	9.3	9.7	10.4	11.7	14.1

Problem 17: The figure below depicts lactate concentrations in 99 patients.



The mean and SD of these measurements are 2.74 mmol/l and 2.60 mmol/l, respectively. Calculate the 95% reference range.

Problem 18: Urinary nicotine excretion ($\mu\text{g}/\text{mg}$ creatinine) related to number of cigarettes smoked daily:

Cigarettes smoked per day	No. in group	Mean	SE
1-9	25	0.31	0.08
10-19	57	0.42	0.10
20-29	99	0.87	0.19
30-39	38	1.03	0.25
>40	28	1.56	0.57
Unspecified	25	0.56	0.16

Calculate the standard deviations and comment on the skewness of the data.

Problem 19: Calculate the standard error and 95% confidence interval for the following data:

5.4	8.2	9.3	9.9	10.5	11.9
6.4	8.3	9.4	9.9	10.5	12.3
6.4	8.3	9.4	9.9	10.6	12.6
7.0	8.6	9.4	10.1	10.8	12.7
7.1	8.8	9.4	10.3	10.8	13.0
7.3	8.9	9.5	10.3	11.3	13.3
7.7	9.1	9.7	10.4	11.7	14.0
8.1	9.3	9.7	10.4	11.7	14.1

Compare the latter with the 95% reference range. If the sample size was 480 instead

of 48, how would the standard error and 95% confidence interval change?

Problem 20: Calculate the 99% confidence interval for the data given in Problem 19. How does the 95% confidence interval (obtained in Problem 19) differ from the 99% confidence interval?

Problem 21: Calculate the 95% confidence interval for the following data:

5.4	8.2	9.3	9.9	10.5	11.9
6.4	8.3	9.4	9.9	10.5	12.3
6.4	8.3	9.4	9.9	10.6	12.6
7.0	8.6	9.4	10.1	10.8	12.7
7.1	8.8	9.4	10.3	10.8	13.0
7.3	8.9	9.5	10.3	11.3	13.3
7.7	9.1	9.7	10.4	11.7	14.0
8.1	9.3	9.7	10.4	11.7	14.1

If the sample size was 10 instead of 48, which distribution would have to be used to calculate the 95% confidence interval? Compare the result with the 95% confidence interval calculated above.

Problem 22: A professor's exam scores are approximately normally distributed with a mean score of 80 and standard deviation 5. What is the probability that an average of three scores is 82 or less?

Problem 23: List the possible outcomes that could occur if an unbiased coin is flipped twice. Tabulate and plot the probabilities of getting 0, 1 and 2 heads.

Problem 24: A list is obtained of birthweights from 100 consecutive, full-term, live-born deliveries from the maternity ward of a hospital in a low-socio-economic status area. The mean birthweight is found to be 115 oz with a sample standard deviation of 24 oz. If it is known from nationwide surveys based on millions of deliveries that the mean birthweight is 120 oz, set up null and alternate hypotheses to test whether the underlying mean birthweight from this hospital is lower than the national average.

Problem 25: Suppose the average cholesterol level in children is 175 mg/dL. A group of men who have died from heart disease within the past year are identified, and the cholesterol levels of their offspring are measured. Two hypotheses are considered: (1) the average cholesterol level of these children is 175 mg/dL, and (2) the average cholesterol level of these children is >175 mg/dL. State the type I and type II errors that can be committed when testing the hypotheses.

Problem 26: A list is obtained of birthweights from 100 consecutive, full-term, live-born deliveries from the maternity ward of a hospital in a low-socio-economic status area. The mean birthweight is found to be 115 oz with a sample standard deviation of 24 oz. If it is known from nationwide surveys based on millions of deliveries that the mean birthweight is 120 oz, use the t test to test the hypothesis $H_0 : \mu=120$ vs. $H_1 : \mu<120$ using significance levels of 0.05 and 0.01. (Given: $t_{99,0.05} = -1.66$ and $t_{99,0.01} = -2.36$)

Problem 27: Suppose the mean cholesterol level of 10 children whose fathers died from heart disease is 200 mg/dL and the sample standard deviation is 50 mg/dL. Test the hypothesis that the mean cholesterol level is higher in this group than in the general population, which is 175 mg/dL. (Given: $t_{9,0.90} = 1.383$ and $t_{9,0.95} = 1.833$)

Problem 28: The mean fasting serum-cholesterol levels among 100 recent female Asian immigrants aged 21–40 to the United States was found to be 181.52 mg/dL with a standard deviation of 40 mg/dL. The cholesterol levels in women in the same age group in the US are approximately normally distributed with a mean of 190 mg/dL. Test the hypothesis at a 5% level of significance that the mean cholesterol level of recent female Asian immigrants is different from the mean of the general U.S. population. (Given: $t_{99,0.975} < 2.000$)

Problem 29: Suppose the standard deviation in Problem 28 is known to be 40, and the sample size is 200. Assess the significance of the results.

Problem 30: The mean infarct size in untreated patients is 25. In 8 patients treated with a drug the mean infarct size is 16 with a standard deviation of 10. Is the drug effective in reducing infarct size? (Given: $t_{7,0.025} = -2.365$ and $t_{7,0.01} = -2.998$)

Problem 31: A list is obtained of birthweights from 10 consecutive, full-term, live-born deliveries from the maternity ward of a hospital in a low-socio-economic status area. The mean birthweight is found to be 110 oz with a sample standard deviation of 24 oz. If it is known from nationwide surveys based on millions of deliveries that the mean birthweight is 120 oz, assess the results of the study. (Given: $t_{9,0.15} = -1.100$ and $t_{9,0.10} = -1.383$)

Problem 32: The mean fasting serum-cholesterol levels among 200 recent female Asian immigrants aged 21–40 to the United States was found to be 185 mg/dL with a standard deviation of 40 mg/dL. The cholesterol levels in women in the same age group in the US are approximately normally distributed with a mean of 190 mg/dL. Calculate the 95% confidence interval and assess the significance of the results when compared with the hypothesis-testing approach.

Problem 33: A list is obtained of birthweights from 100 consecutive, full-term, live-born deliveries from the maternity ward of a hospital in a low-socio-economic status area. Suppose that $\mu_0 = 120$ oz, $\mu_1 = 115$ oz, $\sigma = 24$, $\alpha = 0.05$, $1-\beta = 0.80$. Compute the appropriate sample size needed to conduct a one-tailed test. (Given: $z_{0.80} = 0.84$ and $z_{0.95} = 1.645$)

Problem 34: Compute the sample size for the birthweight data in Problem 33 if $\mu_1 = 110$ oz rather than 115 oz.

Problem 35: A new drug for the treatment of patients with angina is to be tested. The effect this drug will have on the heart rate is unknown. Suppose the change in heart rate after 48 hours is known to have a standard deviation of 10 beats per minute. If at least 80% power is desired for detecting a significant difference at a significance level of 5% if the effect of the drug is to change mean heart rate by 5 beats per minute over 48 hours in either direction, how many patients should be enrolled in such a study? (Given: $z_{0.80} = 0.84$ and $z_{0.975} = 1.96$)

Problem 36: Central venous oxygen saturation (%) in 10 patients on admission and 6 hours after admission to an intensive care unit:

Subject	On admission	6 h after admission
1	39.7	52.9
2	59.1	56.7
3	56.1	61.9
4	57.7	71.4
5	60.6	67.7
6	37.8	50.0
7	58.2	60.7
8	33.6	51.3
9	56.0	59.5
10	65.3	59.8

Test the hypothesis that admission to ICU and subsequent treatment may change central venous oxygen saturation beyond the level expected by chance.

Problem 37: Mean arterial blood pressure measurements from a trial that compared early goal-directed therapy with standard therapy in the treatment of severe sepsis and septic shock are given below:

	Standard therapy	Early goal-directed therapy
Number of patients	119	117
Mean	81	95
Standard deviation	18	19

Test the hypothesis that the difference in mean arterial pressure between the two groups would be observed just by chance.